Section 2 (Take Home Section)

Mark: /12

You will complete the experiment as we did in class, but your experiment should be without replacement. The aim of the experiment is to compare theoretical and experimental probability.

1. How many times do you think you should complete the experiment? Give reasons for your answer.

(2 marks)

~50 or more.

to approach the theoretical probability.

2. Complete a frequency table to record your results:

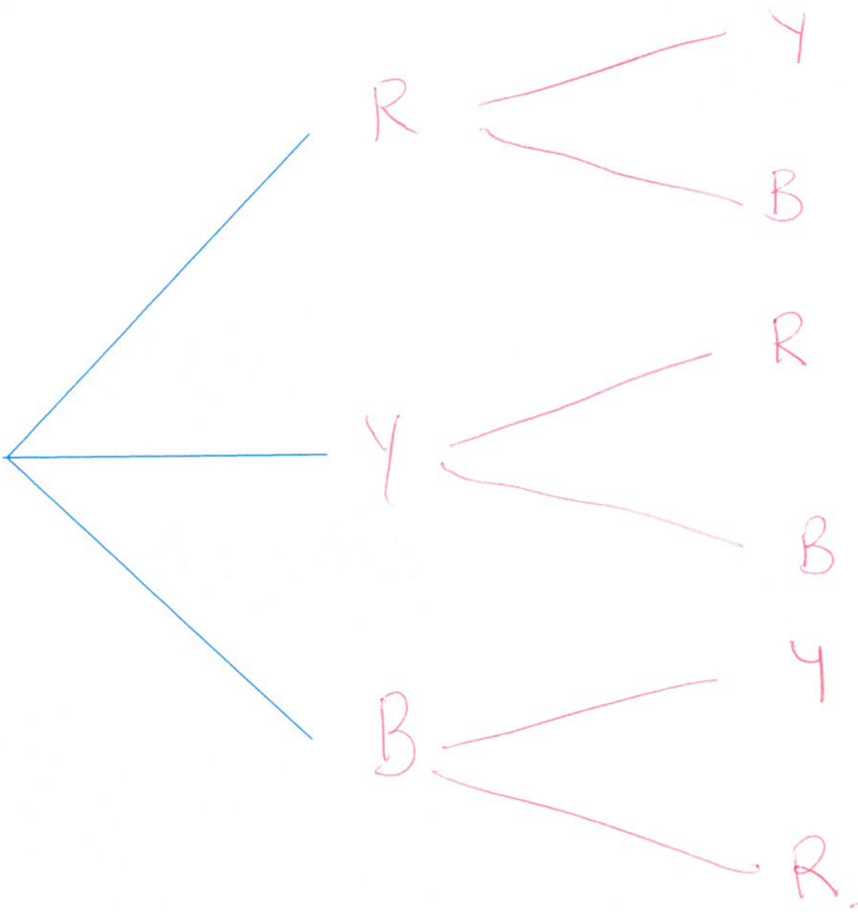
(4 marks)

(The following table has been provided as a suggestion. Not all rows may be required.)

First	Second	Tally	Frequency	Relative Frequency
Blue	Red			
Blue	Yellow			
Blue	Blue	0	0	0
Red	Blue			
Red	Yellow			
Red	Red	0	0	0
Yellow	Blue			
Yellow	Red			
Yellow	Yellow	0.	0	0
		Total		

3. Draw a tree diagram to demonstrate the theoretical probability for this experiment.

(4 marks)



Outcomes.

RY	$\frac{1}{6}$
RB	$\frac{1}{6}$
YR	$\frac{1}{6}$
YB	$\frac{1}{6}$
BY	$\frac{1}{6}$
BR	$\frac{1}{6}$

4. Is there a difference between the theoretical and relative frequency (experimental probability) for the outcomes? Explain why/why not.

(2 marks)

Probably. because the experiment is a chance event.